

Name: Varia Neel Jitesh

Enrollment No: 250447003019

Network Security And Forensics

Lab Session – 3

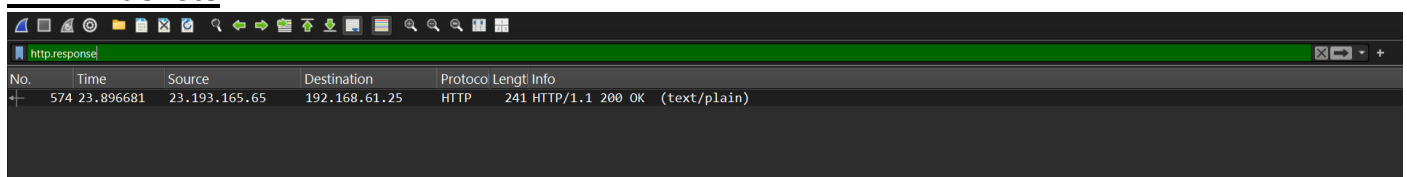
Perform the depth protocol analysis by Wireshark.

Set up Wireshark to capture packets on a network interface.

- Capture and analyze various types of traffic, such as HTTP, DNS, and FTP.
- Examine packet headers, source and destination IP addresses, port numbers, and protocols.
- Use display filters to focus on specific types of traffic
- Experiment with different display filter expressions.
- Explore and understand color-coded packet markings.
- Analyses the handshake process in
- Investigate DNS query and response packets.
- Examine HTTP requests and responses, including headers and content. Packet Filtering and Display Filters:
- Use Wireshark to capture network traffic in a specific scenario (e.g., browsing a website, downloading a file).
- Apply display filters in Wireshark to isolate specific types of packets (e.g., HTTP, TCP).
- Experiment with different display filters and observe how they affect the captured traffic view.
- How can display filters help in focusing on relevant packets during network analysis?

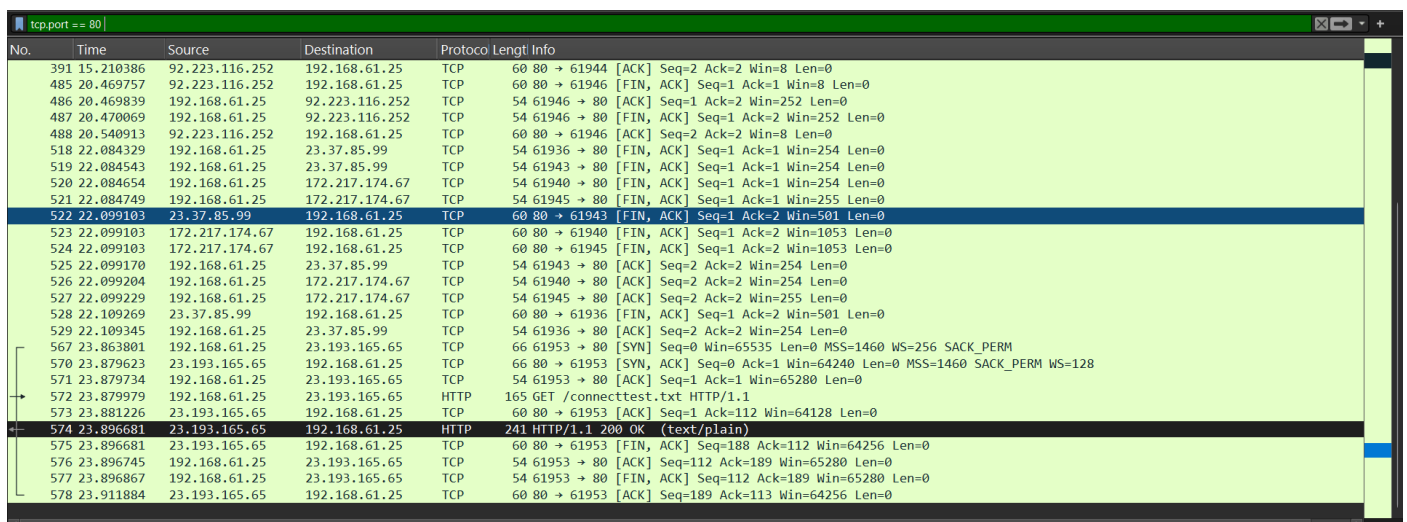
Capture and analyze various types of traffic, such as HTTP, DNS, and FTP.

HTTP Packets



The image shows a Wireshark packet capture window with the display filter 'http.response'. The packet list shows a single packet at time 574.23.896681, source 23.193.165.65, destination 192.168.61.25, protocol HTTP, length 241, and info 'HTTP/1.1 200 OK (text/plain)'.

No.	Time	Source	Destination	Protocol	Length	Info
574	23.896681	23.193.165.65	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)



The image shows a Wireshark packet capture window with the display filter 'tcp.port == 80'. The packet list shows a sequence of packets from time 391 to 578. The packets include TCP handshakes, data transfers, and an HTTP GET request and response.

No.	Time	Source	Destination	Protocol	Length	Info
391	15.210386	92.223.116.252	192.168.61.25	TCP	60	80 → 61944 [ACK] Seq=2 Ack=2 Win=8 Len=0
485	20.469757	92.223.116.252	192.168.61.25	TCP	60	80 → 61946 [FIN, ACK] Seq=1 Ack=1 Win=8 Len=0
486	20.469839	192.168.61.25	92.223.116.252	TCP	54	61946 → 80 [ACK] Seq=1 Ack=2 Win=252 Len=0
487	20.470069	192.168.61.25	92.223.116.252	TCP	54	61946 → 80 [FIN, ACK] Seq=1 Ack=2 Win=252 Len=0
488	20.540913	92.223.116.252	192.168.61.25	TCP	60	80 → 61946 [ACK] Seq=2 Ack=2 Win=8 Len=0
518	22.084329	192.168.61.25	23.37.85.99	TCP	54	61936 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
519	22.084543	192.168.61.25	23.37.85.99	TCP	54	61943 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
520	22.084654	192.168.61.25	172.217.174.67	TCP	54	61940 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
521	22.084749	192.168.61.25	172.217.174.67	TCP	54	61945 → 80 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0
522	22.099103	23.37.85.99	192.168.61.25	TCP	60	80 → 61943 [FIN, ACK] Seq=1 Ack=2 Win=501 Len=0
523	22.099103	172.217.174.67	192.168.61.25	TCP	60	80 → 61940 [FIN, ACK] Seq=1 Ack=2 Win=1053 Len=0
524	22.099103	172.217.174.67	192.168.61.25	TCP	60	80 → 61945 [FIN, ACK] Seq=1 Ack=2 Win=1053 Len=0
525	22.099170	192.168.61.25	23.37.85.99	TCP	54	61943 → 80 [ACK] Seq=2 Ack=2 Win=254 Len=0
526	22.099204	192.168.61.25	172.217.174.67	TCP	54	61940 → 80 [ACK] Seq=2 Ack=2 Win=254 Len=0
527	22.099229	192.168.61.25	172.217.174.67	TCP	54	61945 → 80 [ACK] Seq=2 Ack=2 Win=255 Len=0
528	22.109269	23.37.85.99	192.168.61.25	TCP	60	80 → 61936 [FIN, ACK] Seq=1 Ack=2 Win=501 Len=0
529	22.109345	192.168.61.25	23.37.85.99	TCP	54	61936 → 80 [ACK] Seq=2 Ack=2 Win=254 Len=0
567	23.863801	192.168.61.25	23.193.165.65	TCP	66	61953 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
570	23.879623	23.193.165.65	192.168.61.25	TCP	66	80 → 61953 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
571	23.879734	192.168.61.25	23.193.165.65	TCP	54	61953 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
572	23.879779	192.168.61.25	23.193.165.65	HTTP	165	GET /connecttest.txt HTTP/1.1
573	23.881226	23.193.165.65	192.168.61.25	TCP	60	80 → 61953 [ACK] Seq=1 Ack=112 Win=64128 Len=0
574	23.896681	23.193.165.65	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)
575	23.896681	23.193.165.65	192.168.61.25	TCP	60	80 → 61953 [FIN, ACK] Seq=188 Ack=112 Win=64256 Len=0
576	23.896745	192.168.61.25	23.193.165.65	TCP	54	61953 → 80 [ACK] Seq=112 Ack=189 Win=65280 Len=0
577	23.896867	192.168.61.25	23.193.165.65	TCP	54	61953 → 80 [FIN, ACK] Seq=112 Ack=189 Win=65280 Len=0
578	23.911884	23.193.165.65	192.168.61.25	TCP	60	80 → 61953 [ACK] Seq=189 Ack=113 Win=64256 Len=0

DNS

udp.port == 53					
No.	Time	Source	Destination	Protocol	Length Info
565	23.862084	192.168.61.25	192.168.60.1	DNS	83 Standard query 0x54b3 A www.msftconnecttest.com
566	23.862991	192.168.60.1	192.168.61.25	DNS	227 Standard query response 0x54b3 A www.msftconnecttest.com CNAME ncsi-geo.trafficmanager.net CNAME www.msftncsi.com.e...

dns.flags.response == 1					
No.	Time	Source	Destination	Protocol	Length Info
53	2.089074	192.168.63.108	224.0.0.251	MDNS	318 Standard query response 0x0000 PTR COMP-04._dosvc._tcp.local SRV 0 0 7680 COMP-04.local TXT
77	2.890238	192.168.63.108	224.0.0.251	MDNS	380 Standard query response 0x0000 PTR, cache flush COMP-04._dosvc._tcp.local SRV, cache flush 0 0 7680 COMP-04.local
78	2.896182	192.168.63.108	224.0.0.251	MDNS	324 Standard query response 0x0000 SRV, cache flush 0 0 7680 COMP-04.local TXT, cache flush A, cache flush 192.168.
80	3.061673	192.168.60.171	224.0.0.251	MDNS	364 Standard query response 0x0000 PTR DESKTOP-BLP031R._dosvc._tcp.local SRV 0 0 7680 DESKTOP-BLP031R.local TXT
81	3.062584	fe80::69a1:d28a:...	ff02::fb	MDNS	384 Standard query response 0x0000 PTR DESKTOP-BLP031R._dosvc._tcp.local SRV 0 0 7680 DESKTOP-BLP031R.local TXT
84	3.066224	192.168.63.107	224.0.0.251	MDNS	364 Standard query response 0x0000 PTR DESKTOP-BLP031R._dosvc._tcp.local SRV 0 0 7680 DESKTOP-BLP031R.local TXT
85	3.066224	fe80::298a:bb2:c...	ff02::fb	MDNS	384 Standard query response 0x0000 PTR DESKTOP-BLP031R._dosvc._tcp.local SRV 0 0 7680 DESKTOP-BLP031R.local TXT
115	3.830675	192.168.60.171	224.0.0.251	MDNS	485 Standard query response 0x0000 PTR, cache flush DESKTOP-BLP031R._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
118	3.831987	fe80::69a1:d28a:...	ff02::fb	MDNS	505 Standard query response 0x0000 PTR, cache flush DESKTOP-BLP031R._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
119	3.833008	192.168.60.171	224.0.0.251	MDNS	421 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-BLP031R.local TXT, cache flush A, cache flush
120	3.833218	fe80::69a1:d28a:...	ff02::fb	MDNS	441 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-BLP031R.local TXT, cache flush A, cache flush
121	3.841594	192.168.63.107	224.0.0.251	MDNS	429 Standard query response 0x0000 PTR, cache flush DESKTOP-BLP031R._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
122	3.841594	fe80::298a:bb2:c...	ff02::fb	MDNS	449 Standard query response 0x0000 PTR, cache flush DESKTOP-BLP031R._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
125	3.848339	192.168.63.107	224.0.0.251	MDNS	365 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-BLP031R.local TXT, cache flush A, cache flush
126	3.854370	fe80::298a:bb2:c...	ff02::fb	MDNS	385 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-BLP031R.local TXT, cache flush A, cache flush
162	4.918200	192.168.61.40	224.0.0.251	MDNS	804 Standard query response 0x0000 PTR DESKTOP-K3MM1JF._dosvc._tcp.local SRV 0 0 7680 DESKTOP-K3MM1JF.local TXT
163	4.920921	fe80::e45f:550a:...	ff02::fb	MDNS	824 Standard query response 0x0000 PTR DESKTOP-K3MM1JF._dosvc._tcp.local SRV 0 0 7680 DESKTOP-K3MM1JF.local TXT
195	5.702052	192.168.61.40	224.0.0.251	MDNS	869 Standard query response 0x0000 PTR, cache flush DESKTOP-K3MM1JF._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
196	5.704060	fe80::e45f:550a:...	ff02::fb	MDNS	889 Standard query response 0x0000 PTR, cache flush DESKTOP-K3MM1JF._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
197	5.705231	192.168.61.40	224.0.0.251	MDNS	805 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-K3MM1JF.local TXT, cache flush A, cache flush
198	5.706464	fe80::e45f:550a:...	ff02::fb	MDNS	825 Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-K3MM1JF.local TXT, cache flush A, cache flush
290	10.292068	192.168.61.8	224.0.0.251	MDNS	894 Standard query response 0x0000 PTR DESKTOP-HR5QVQE._dosvc._tcp.local SRV 0 0 7680 DESKTOP-HR5QVQE.local TXT
291	10.292068	fe80::3de0:cdca:...	ff02::fb	MDNS	914 Standard query response 0x0000 PTR DESKTOP-HR5QVQE._dosvc._tcp.local SRV 0 0 7680 DESKTOP-HR5QVQE.local TXT
310	11.075314	192.168.61.8	224.0.0.251	MDNS	959 Standard query response 0x0000 PTR DESKTOP-HR5QVQE._dosvc._tcp.local SRV 0 0 7680 DESKTOP-HR5QVQE.local TXT A 1
312	11.076225	fe80::3de0:cdca:...	ff02::fb	MDNS	979 Standard query response 0x0000 PTR DESKTOP-HR5QVQE._dosvc._tcp.local SRV 0 0 7680 DESKTOP-HR5QVQE.local TXT A 1
313	11.076225	192.168.61.8	224.0.0.251	MDNS	895 Standard query response 0x0000 SRV 0 0 7680 DESKTOP-HR5QVQE.local TXT A 192.168.61.8 AAAA fe80::3de0:cdca:6119:
314	11.076780	fe80::3de0:cdca:...	ff02::fb	MDNS	915 Standard query response 0x0000 SRV 0 0 7680 DESKTOP-HR5QVQE.local TXT A 192.168.61.8 AAAA fe80::3de0:cdca:6119:
550	23.166993	192.168.60.110	224.0.0.251	MDNS	408 Standard query response 0x0000 PTR DESKTOP-UM2V9T3._dosvc._tcp.local SRV 0 0 7680 DESKTOP-UM2V9T3.local TXT

Layer	Header Type	Purpose
Layer 2	Ethernet Header	MAC addressing
Layer 3	IP Header	Logical addressing
Layer 4	TCP / UDP Header	Port numbers, reliability
Layer 7	Application Header	Protocol-specific data

Packet Headers

▶ Frame 123: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface
▼ Ethernet II, Src: Sophos_fc:00:01 (c8:4f:86:fc:00:01), Dst: ASUSTekCOMPU_d4:33:45 (60:cf:8
Destination: ASUSTekCOMPU_d4:33:45 (60:cf:84:d4:33:45)
.... ..0. = LG bit: Globally unique address (factory default)
.... ...0 = IG bit: Individual address (unicast)
Source: Sophos_fc:00:01 (c8:4f:86:fc:00:01)
.... ..0. = LG bit: Globally unique address (factory default)
.... ...0 = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 19]
Padding: 000000000000
▶ Internet Protocol Version 4, Src: 142.251.222.106, Dst: 192.168.61.25
▶ Transmission Control Protocol, Src Port: 443, Dst Port: 58713, Seq: 248, Ack: 36, Len: 0

Examining IP Header

The image shows a Wireshark packet capture of an Ethernet II frame. The packet is 60 bytes on wire (480 bits) and 60 bytes captured (480 bits) on interface eth0. The Ethernet II header shows the source as Sophos_fc:00:01 (c8:4f:86:fc:00:01) and the destination as ASUSTekCOMPU_d4:33:45 (60:cf:00:28:6d:61). The Internet Protocol Version 4 header shows the source as 142.251.222.106 and the destination as 192.168.61.25. The IP header details include: Version: 4, Header Length: 20 bytes (5), Differentiated Services Field: 0x80 (DSCP: CS4, ECN: Not-ECT), Total Length: 40, Identification: 0x6d61 (28001), Flags: 0x0, Fragment Offset: 0, Time to Live: 124, Protocol: TCP (6), Header Checksum: 0x65c7 [validation disabled], [Header checksum status: Unverified], Source Address: 142.251.222.106, Destination Address: 192.168.61.25, [Stream index: 10]. The Transmission Control Protocol header shows: Source Port: 443, Destination Port: 58713, [Stream index: 1], [Stream Packet Number: 4], [Conversation completeness: Incomplete (12)], [TCP Segment Len: 0], Sequence Number: 248 (relative sequence number), Sequence Number (raw): 2634618184, [Next Sequence Number: 248 (relative sequence number)], Acknowledgment Number: 36 (relative ack number), Acknowledgment number (raw): 643468919, 0101 = Header Length: 20 bytes (5), Flags: 0x010 (ACK), Window: 993, [Calculated window size: 993], [Window size scaling factor: -1 (unknown)]. The packet details pane is expanded to show the IP header, and the packet bytes pane shows the raw data in hexadecimal and ASCII.

Examining Application Layer Protocols

The image shows a Wireshark packet capture of an HTTP response. The packet is 241 bytes long. The Ethernet II header shows the source as 23.193.165.65 and the destination as 192.168.61.25. The Internet Protocol Version 4 header shows the source as 23.193.165.65 and the destination as 192.168.61.25. The Transmission Control Protocol header shows: Source Port: 80, Destination Port: 61953, [Stream index: 14], [Stream Packet Number: 6], [Conversation completeness: Complete, WITH_DATA (31)], [TCP Segment Len: 187], Sequence Number: 1 (relative sequence number), Sequence Number (raw): 2048403035, [Next Sequence Number: 188 (relative sequence number)], Acknowledgment Number: 112 (relative ack number), Acknowledgment number (raw): 2602890469, 0101 = Header Length: 20 bytes (5), Flags: 0x018 (PSH, ACK), Window: 502, [Calculated window size: 64256], [Window size scaling factor: 128], Checksum: 0x1c70 [unverified], [Checksum Status: Unverified], Urgent Pointer: 0, [Timestamps], [SEQ/ACK analysis], [Client Contiguous Streams: 1], [Server Contiguous Streams: 1], TCP payload (187 bytes). The Hypertext Transfer Protocol header shows: Line-based text data: text/plain (1 lines). The packet details pane is expanded to show the HTTP header, and the packet bytes pane shows the raw data in hexadecimal and ASCII.

Use display filters to focus on specific types of traffic

Specific HTTP Filters

http.request

http.request						
No.	Time	Source	Destination	Protocol	Length	Info
401	9.139974	192.168.61.171	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
404	9.170879	192.168.61.171	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
440	10.012010	192.168.61.25	23.193.165.65	HTTP	165	GET /connecttest.txt HTTP/1.1
634	15.491458	192.168.63.59	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
635	15.523481	192.168.63.59	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
720	18.531515	192.168.63.59	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
722	18.563947	192.168.63.59	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
1307	40.087442	192.168.61.25	23.193.165.43	HTTP	165	GET /connecttest.txt HTTP/1.1
2457	66.971025	192.168.60.229	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
2606	69.982979	192.168.60.229	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
2618	70.176271	192.168.61.25	23.193.165.43	HTTP	165	GET /connecttest.txt HTTP/1.1
2830	72.994757	192.168.60.229	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
2867	73.755680	192.168.61.145	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
2868	73.759774	192.168.61.246	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
2893	74.244863	192.168.61.9	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
2946	75.253453	192.168.61.9	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
3087	76.260570	192.168.61.9	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
3181	76.763267	192.168.61.145	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
3218	76.958416	192.168.61.246	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
3249	77.273321	192.168.61.9	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
16270	85.885043	192.168.61.171	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
16274	85.916310	192.168.61.171	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
16347	88.932262	192.168.61.171	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
16349	88.953229	192.168.61.171	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
16713	100.304739	192.168.61.25	23.193.165.65	HTTP	165	GET /connecttest.txt HTTP/1.1

http.response

http.response						
No.	Time	Source	Destination	Protocol	Length	Info
443	10.044141	23.193.165.65	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)
1309	40.105046	23.193.165.43	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)
2628	70.207598	23.193.165.43	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)
16715	100.350816	23.193.165.65	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)

http.request.method == "GET"

http.request.method == "GET"						
No.	Time	Source	Destination	Protocol	Length	Info
440	10.012010	192.168.61.25	23.193.165.65	HTTP	165	GET /connecttest.txt HTTP/1.1
1307	40.087442	192.168.61.25	23.193.165.43	HTTP	165	GET /connecttest.txt HTTP/1.1
2618	70.176271	192.168.61.25	23.193.165.43	HTTP	165	GET /connecttest.txt HTTP/1.1
16713	100.304739	192.168.61.25	23.193.165.65	HTTP	165	GET /connecttest.txt HTTP/1.1

tcp.port == 80

tcp.port == 80						
No.	Time	Source	Destination	Protocol	Length	Info
437	9.980990	192.168.61.25	23.193.165.65	TCP	66	57376 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
438	10.011355	23.193.165.65	192.168.61.25	TCP	66	80 → 57376 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
439	10.011493	192.168.61.25	23.193.165.65	TCP	54	57376 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
440	10.012010	192.168.61.25	23.193.165.65	HTTP	165	GET /connecttest.txt HTTP/1.1
441	10.013296	23.193.165.65	192.168.61.25	TCP	60	80 → 57376 [ACK] Seq=1 Ack=112 Win=64128 Len=0
443	10.044141	23.193.165.65	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)
444	10.044141	23.193.165.65	192.168.61.25	TCP	60	80 → 57376 [FIN, ACK] Seq=188 Ack=112 Win=64256 Len=0
445	10.044244	192.168.61.25	23.193.165.65	TCP	54	57376 → 80 [ACK] Seq=112 Ack=189 Win=65280 Len=0
446	10.044420	192.168.61.25	23.193.165.65	TCP	54	57376 → 80 [FIN, ACK] Seq=112 Ack=189 Win=65280 Len=0
447	10.073728	23.193.165.65	192.168.61.25	TCP	60	80 → 57376 [ACK] Seq=189 Ack=113 Win=64256 Len=0
1304	40.071285	192.168.61.25	23.193.165.43	TCP	66	60392 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1305	40.087204	23.193.165.43	192.168.61.25	TCP	66	80 → 60392 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1306	40.087307	192.168.61.25	23.193.165.43	TCP	54	60392 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
1307	40.087442	192.168.61.25	23.193.165.43	HTTP	165	GET /connecttest.txt HTTP/1.1
1308	40.088494	23.193.165.43	192.168.61.25	TCP	60	80 → 60392 [ACK] Seq=1 Ack=112 Win=64128 Len=0
1309	40.105046	23.193.165.43	192.168.61.25	HTTP	241	HTTP/1.1 200 OK (text/plain)
1310	40.105046	23.193.165.43	192.168.61.25	TCP	60	80 → 60392 [FIN, ACK] Seq=188 Ack=112 Win=64256 Len=0
1311	40.105155	192.168.61.25	23.193.165.43	TCP	54	60392 → 80 [ACK] Seq=112 Ack=189 Win=65280 Len=0
1312	40.105287	192.168.61.25	23.193.165.43	TCP	54	60392 → 80 [FIN, ACK] Seq=112 Ack=189 Win=65280 Len=0
1314	40.120599	23.193.165.43	192.168.61.25	TCP	60	80 → 60392 [ACK] Seq=189 Ack=113 Win=64256 Len=0
2611	70.145510	192.168.61.25	23.193.165.43	TCP	66	57607 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2616	70.175878	23.193.165.43	192.168.61.25	TCP	66	80 → 57607 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
2617	70.176019	192.168.61.25	23.193.165.43	TCP	54	57607 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
2618	70.176271	192.168.61.25	23.193.165.43	HTTP	165	GET /connecttest.txt HTTP/1.1
2619	70.177491	23.193.165.43	192.168.61.25	TCP	60	80 → 57607 [ACK] Seq=1 Ack=112 Win=64128 Len=0

Frame 437: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0000 c8 4f 86 fc 00 01 60 cf 84 d4 33 45 08 00 45 00 00 3E E

Used to analyze web browsing activity.

udp.port == 53

No.	Time	Source	Destination	Protocol	Length	Info
47	1.911025	192.168.61.25	192.168.60.1	DNS	78	Standard query 0x0fc1 A www.howtoforge.com
51	1.940055	192.168.61.25	8.8.8.8	DNS	78	Standard query 0x0fc1 A www.howtoforge.com
54	1.958983	192.168.60.1	192.168.61.25	DNS	126	Standard query response 0x0fc1 A www.howtoforge.com A 172.67.68.93 A 104.26.3.165 A 104.26.2.165
55	1.960686	8.8.8.8	192.168.61.25	DNS	126	Standard query response 0x0fc1 A www.howtoforge.com A 104.26.3.165 A 172.67.68.93 A 104.26.2.165
435	9.978290	192.168.61.25	192.168.60.1	DNS	83	Standard query 0x8812 A www.msftconnecttest.com
436	9.979217	192.168.60.1	192.168.61.25	DNS	227	Standard query response 0x8812 A www.msftconnecttest.com CNAME ncsi-geo.trafficmanager.net CNAME www.msftncsi.c
815	21.658500	192.168.61.25	192.168.60.1	DNS	75	Standard query 0x618d A wpad.nfsu.ac.in
816	21.659538	192.168.60.1	192.168.61.25	DNS	75	Standard query response 0x618d No such name A wpad.nfsu.ac.in
817	21.663200	192.168.61.25	192.168.60.1	DNS	75	Standard query 0x242e A wpad.nfsu.ac.in
818	21.664133	192.168.60.1	192.168.61.25	DNS	75	Standard query response 0x242e No such name A wpad.nfsu.ac.in
1284	39.955367	192.168.61.25	192.168.60.1	DNS	70	Standard query 0xfa60 HTTPS dns.google
1285	39.955526	192.168.61.25	192.168.60.1	DNS	70	Standard query 0xecdf A dns.google
1286	39.956359	192.168.60.1	192.168.61.25	DNS	70	Standard query response 0xfa60 HTTPS dns.google
1287	39.956359	192.168.60.1	192.168.61.25	DNS	102	Standard query response 0xecdf A dns.google A 8.8.8.8 A 8.8.4.4
1302	40.069290	192.168.61.25	192.168.60.1	DNS	83	Standard query 0xb12b A www.msftconnecttest.com
1303	40.070307	192.168.60.1	192.168.61.25	DNS	227	Standard query response 0xb12b A www.msftconnecttest.com CNAME ncsi-geo.trafficmanager.net CNAME www.msftncsi.c
1800	56.722686	192.168.61.25	192.168.60.1	DNS	98	Standard query 0x432e A inputsuggestions.msdxcdn.microsoft.com
1803	56.753844	192.168.60.1	192.168.61.25	DNS	145	Standard query response 0x432e No such name A inputsuggestions.msdxcdn.microsoft.com CNAME msdx-cdn-y86ykn8r9v.
1805	56.753969	192.168.61.25	8.8.8.8	DNS	98	Standard query 0x432e A inputsuggestions.msdxcdn.microsoft.com
1809	56.783977	8.8.8.8	192.168.61.25	DNS	205	Standard query response 0x432e No such name A inputsuggestions.msdxcdn.microsoft.com CNAME msdx-cdn-y86ykn8r9v.
1889	60.010110	192.168.61.25	192.168.60.1	DNS	76	Standard query 0x4603 HTTPS web.whatsapp.com
1890	60.010256	192.168.61.25	192.168.60.1	DNS	76	Standard query 0xb062 A web.whatsapp.com
1891	60.010996	192.168.61.25	192.168.60.1	DNS	89	Standard query 0x8c36 HTTPS _5222._https.web.whatsapp.com
1892	60.011117	192.168.60.1	192.168.61.25	DNS	113	Standard query response 0x4603 HTTPS web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net
1893	60.011117	192.168.60.1	192.168.61.25	DNS	129	Standard query response 0xb062 A web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net A 57.144.43.32

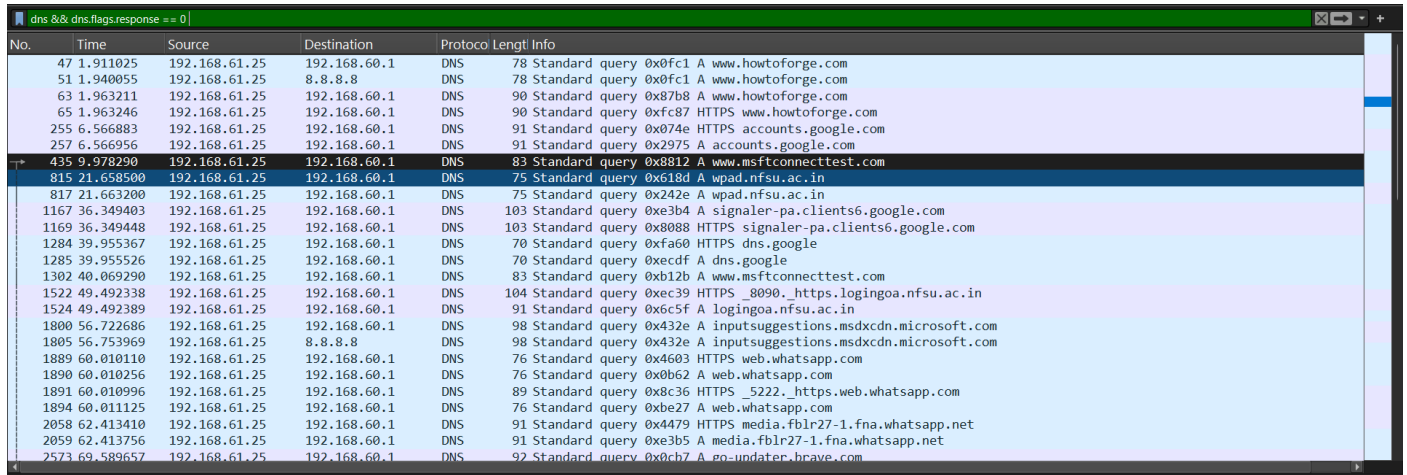
dns.qry.name

No.	Time	Source	Destination	Protocol	Length	Info
266	6.612345	192.168.60.1	192.168.61.25	DNS	93	Standard query response 0x074e HTTPS accounts.google.com
380	8.925585	fe80::4f5b:179d...	ff02::1:3	LLMNR	95	Standard query 0xc252 ANY DESKTOP-7ASOG1U
381	8.925585	192.168.61.21	224.0.0.252	LLMNR	75	Standard query 0xc252 ANY DESKTOP-7ASOG1U
389	8.931698	fe80::82b7:87c3...	ff02::1:3	LLMNR	95	Standard query 0xc252 ANY DESKTOP-7ASOG1U
390	8.931698	192.168.61.171	224.0.0.252	LLMNR	75	Standard query 0xc252 ANY DESKTOP-7ASOG1U
408	9.346863	fe80::82b7:87c3...	ff02::1:3	LLMNR	95	Standard query 0xc252 ANY DESKTOP-7ASOG1U
409	9.346863	192.168.61.171	224.0.0.252	LLMNR	75	Standard query 0xc252 ANY DESKTOP-7ASOG1U
424	9.765898	fe80::82b7:87c3...	ff02::1:3	LLMNR	95	Standard query 0x122b ANY DESKTOP-7ASOG1U
425	9.765898	192.168.61.171	224.0.0.252	LLMNR	75	Standard query 0x122b ANY DESKTOP-7ASOG1U
430	9.881314	192.168.60.105	224.0.0.251	MDNS	85	Standard query 0x4c59 PTR 1.60.168.192.in-addr.arpa, "QM" question
434	9.897127	192.168.60.105	224.0.0.251	MDNS	85	Standard query 0x4c5a PTR 1.60.168.192.in-addr.arpa, "QM" question
435	9.978290	192.168.61.25	192.168.60.1	DNS	83	Standard query 0x8812 A www.msftconnecttest.com
436	9.979217	192.168.60.1	192.168.61.25	DNS	227	Standard query response 0x8812 A www.msftconnecttest.com CNAME ncsi-geo.trafficmanager.net CNAME www.msftncsi.c
628	15.305687	192.168.60.173	224.0.0.251	MDNS	85	Standard query 0x35b7 PTR 192.168.60.1.in-addr.arpa, "QM" question
630	15.316037	192.168.60.173	224.0.0.251	MDNS	85	Standard query 0x35b8 PTR 192.168.60.1.in-addr.arpa, "QM" question
685	17.459841	192.168.63.119	224.0.0.251	MDNS	93	Standard query 0x0000 ANY DESKTOP-GDVN21Q._dosvc._tcp.local, "QM" question
686	17.460604	fe80::6bcb:f8b1...	ff02::fb	MDNS	113	Standard query 0x0000 ANY DESKTOP-GDVN21Q._dosvc._tcp.local, "QM" question
691	17.711667	192.168.63.119	224.0.0.251	MDNS	93	Standard query 0x0000 ANY DESKTOP-GDVN21Q._dosvc._tcp.local, "QM" question
692	17.712288	fe80::6bcb:f8b1...	ff02::fb	MDNS	113	Standard query 0x0000 ANY DESKTOP-GDVN21Q._dosvc._tcp.local, "QM" question
700	17.963393	192.168.63.119	224.0.0.251	MDNS	93	Standard query 0x0000 ANY DESKTOP-GDVN21Q._dosvc._tcp.local, "QM" question
701	17.963393	fe80::6bcb:f8b1...	ff02::fb	MDNS	113	Standard query 0x0000 ANY DESKTOP-GDVN21Q._dosvc._tcp.local, "QM" question
744	19.581802	192.168.63.59	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
745	19.581802	fe80::af6a:4ff3...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
746	19.582202	192.168.63.59	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
747	19.582313	fe80::af6a:4ff3...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question

dns.flags.response == 1

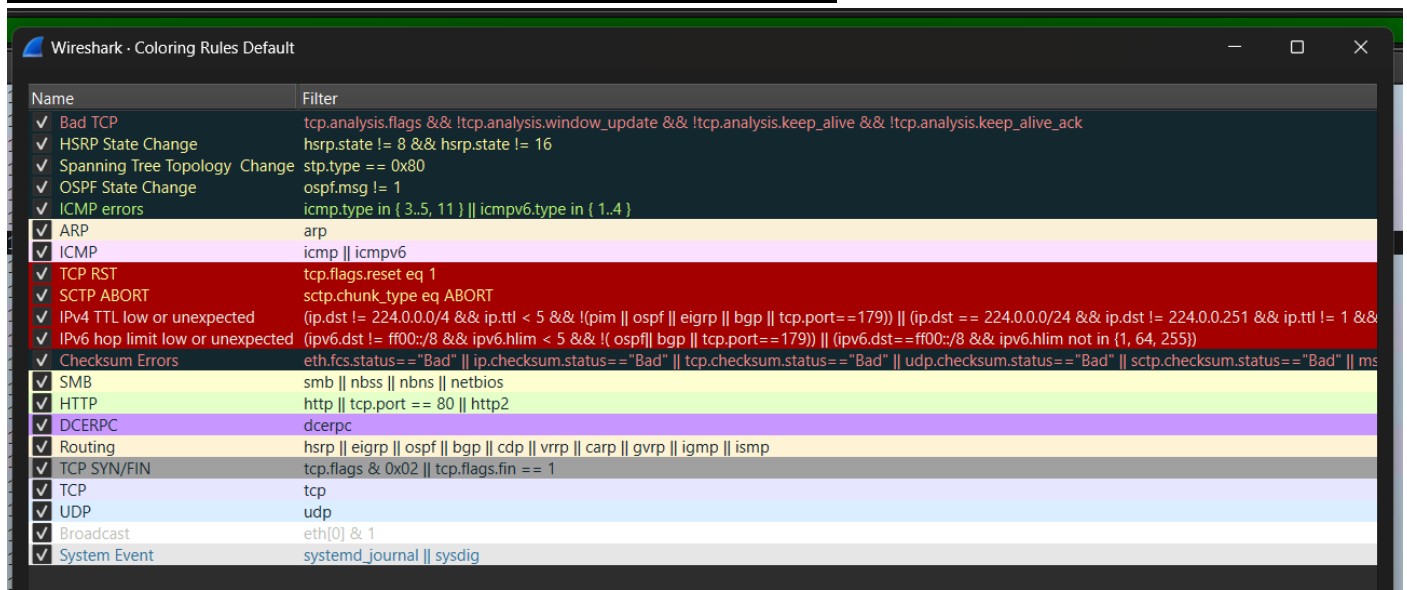
No.	Time	Source	Destination	Protocol	Length	Info
54	1.958983	192.168.60.1	192.168.61.25	DNS	126	Standard query response 0x0fc1 A www.howtoforge.com A 172.67.68.93 A 104.26.3.165 A 104.26.2.165
55	1.960686	8.8.8.8	192.168.61.25	DNS	126	Standard query response 0x0fc1 A www.howtoforge.com A 104.26.3.165 A 172.67.68.93 A 104.26.2.165
70	1.963890	192.168.60.1	192.168.61.25	DNS	140	Standard query response 0x87b8 A www.howtoforge.com A 172.67.68.93 A 104.26.3.165 A 104.26.2.165
98	2.075987	192.168.60.1	192.168.61.25	DNS	185	Standard query response 0xfc87 HTTPS www.howtoforge.com HTTPS
186	4.540900	192.168.61.9	224.0.0.251	MDNS	413	Standard query response 0x0000 PTR HP._dosvc._tcp.local SRV 0 0 7680 HP.local TXT
187	4.540900	fe80::6f8d:93ec...	ff02::fb	MDNS	433	Standard query response 0x0000 PTR HP._dosvc._tcp.local SRV 0 0 7680 HP.local TXT
214	5.332679	192.168.61.9	224.0.0.251	MDNS	521	Standard query response 0x0000 PTR, cache flush HP._dosvc._tcp.local SRV, cache flush 0 0 7680 HP.local TXT, ca
215	5.332834	fe80::6f8d:93ec...	ff02::fb	MDNS	541	Standard query response 0x0000 PTR, cache flush HP._dosvc._tcp.local SRV, cache flush 0 0 7680 HP.local TXT, ca
216	5.333816	192.168.61.9	224.0.0.251	MDNS	470	Standard query response 0x0000 SRV, cache flush 0 0 7680 HP.local TXT, cache flush A, cache flush 192.168.61.9
217	5.333816	fe80::6f8d:93ec...	ff02::fb	MDNS	490	Standard query response 0x0000 SRV, cache flush 0 0 7680 HP.local TXT, cache flush A, cache flush 192.168.61.9
262	6.597149	192.168.60.1	192.168.61.25	DNS	109	Standard query response 0x2975 A accounts.google.com A 192.178.211.84
266	6.612345	192.168.60.1	192.168.61.25	DNS	93	Standard query response 0x074e HTTPS accounts.google.com
436	9.979217	192.168.60.1	192.168.61.25	DNS	227	Standard query response 0x8812 A www.msftconnecttest.com CNAME ncsi-geo.trafficmanager.net CNAME www.msftncsi.c
683	17.456675	192.168.63.119	224.0.0.251	MDNS	320	Standard query response 0x0000 PTR DESKTOP-GDVN21Q._dosvc._tcp.local SRV 0 0 7680 DESKTOP-GDVN21Q.local TXT
684	17.458113	fe80::6bcb:f8b1...	ff02::fb	MDNS	340	Standard query response 0x0000 PTR DESKTOP-GDVN21Q._dosvc._tcp.local SRV 0 0 7680 DESKTOP-GDVN21Q.local TXT
708	18.216258	192.168.63.119	224.0.0.251	MDNS	441	Standard query response 0x0000 PTR, cache flush DESKTOP-GDVN21Q._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
709	18.217552	fe80::6bcb:f8b1...	ff02::fb	MDNS	461	Standard query response 0x0000 PTR, cache flush DESKTOP-GDVN21Q._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
710	18.218858	192.168.63.119	224.0.0.251	MDNS	377	Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-GDVN21Q.local TXT, cache flush A, cache flush
711	18.219646	fe80::6bcb:f8b1...	ff02::fb	MDNS	397	Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-GDVN21Q.local TXT, cache flush A, cache flush
774	20.612865	192.168.60.217	224.0.0.251	MDNS	474	Standard query response 0x0000 PTR DESKTOP-COUVH72._dosvc._tcp.local SRV 0 0 7680 DESKTOP-COUVH72.local TXT
775	20.613902	fe80::7ebd:d5c7...	ff02::fb	MDNS	494	Standard query response 0x0000 PTR DESKTOP-COUVH72._dosvc._tcp.local SRV 0 0 7680 DESKTOP-COUVH72.local TXT
807	21.387921	192.168.60.217	224.0.0.251	MDNS	539	Standard query response 0x0000 PTR, cache flush DESKTOP-COUVH72._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
808	21.388823	fe80::7ebd:d5c7...	ff02::fb	MDNS	559	Standard query response 0x0000 PTR, cache flush DESKTOP-COUVH72._dosvc._tcp.local SRV, cache flush 0 0 7680 DES
809	21.390064	192.168.60.217	224.0.0.251	MDNS	475	Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-COUVH72.local TXT, cache flush A, cache flush
810	21.390064	fe80::7ebd:d5c7...	ff02::fb	MDNS	495	Standard query response 0x0000 SRV, cache flush 0 0 7680 DESKTOP-COUVH72.local TXT, cache flush A, cache flush

dns && dns.flags.response == 0



No.	Time	Source	Destination	Protocol	Length	Info
47	1.911025	192.168.61.25	192.168.60.1	DNS	78	Standard query 0x0fc1 A www.howtoforge.com
51	1.940055	192.168.61.25	8.8.8.8	DNS	78	Standard query 0x0fc1 A www.howtoforge.com
63	1.963211	192.168.61.25	192.168.60.1	DNS	90	Standard query 0x87b8 A www.howtoforge.com
65	1.963246	192.168.61.25	192.168.60.1	DNS	90	Standard query 0xfc87 HTTPS www.howtoforge.com
255	6.566883	192.168.61.25	192.168.60.1	DNS	91	Standard query 0x074e HTTPS accounts.google.com
257	6.566956	192.168.61.25	192.168.60.1	DNS	91	Standard query 0x2975 A accounts.google.com
435	9.978290	192.168.61.25	192.168.60.1	DNS	83	Standard query 0x8812 A www.msftconnecttest.com
815	21.658500	192.168.61.25	192.168.60.1	DNS	75	Standard query 0x618d A wpad.nfsu.ac.in
817	21.663200	192.168.61.25	192.168.60.1	DNS	75	Standard query 0x242e A wpad.nfsu.ac.in
1167	36.349403	192.168.61.25	192.168.60.1	DNS	103	Standard query 0xe3b4 A signaler-pa.clients6.google.com
1169	36.349448	192.168.61.25	192.168.60.1	DNS	103	Standard query 0x8088 HTTPS signaler-pa.clients6.google.com
1284	39.955367	192.168.61.25	192.168.60.1	DNS	70	Standard query 0xfa60 HTTPS dns.google
1285	39.955526	192.168.61.25	192.168.60.1	DNS	70	Standard query 0xecdf A dns.google
1302	40.069290	192.168.61.25	192.168.60.1	DNS	83	Standard query 0xb12b A www.msftconnecttest.com
1522	49.492338	192.168.61.25	192.168.60.1	DNS	104	Standard query 0xec39 HTTPS _8090_.https.logingoa.nfsu.ac.in
1524	49.492389	192.168.61.25	192.168.60.1	DNS	91	Standard query 0xc65f A logingoa.nfsu.ac.in
1800	56.722686	192.168.61.25	192.168.60.1	DNS	98	Standard query 0x432e A inputsuggestions.msdxcdn.microsoft.com
1805	56.753969	192.168.61.25	8.8.8.8	DNS	98	Standard query 0x432e A inputsuggestions.msdxcdn.microsoft.com
1889	60.010110	192.168.61.25	192.168.60.1	DNS	76	Standard query 0x4603 HTTPS web.whatsapp.com
1890	60.010256	192.168.61.25	192.168.60.1	DNS	76	Standard query 0xb62 A web.whatsapp.com
1891	60.010996	192.168.61.25	192.168.60.1	DNS	89	Standard query 0x8c36 HTTPS _5222_.https.web.whatsapp.com
1894	60.011125	192.168.61.25	192.168.60.1	DNS	76	Standard query 0xb627 A web.whatsapp.com
2058	62.413410	192.168.61.25	192.168.60.1	DNS	91	Standard query 0x4479 HTTPS media.fb1r27-1.fna.whatsapp.net
2059	62.413756	192.168.61.25	192.168.60.1	DNS	91	Standard query 0xe3b5 A media.fb1r27-1.fna.whatsapp.net
2573	69.589657	192.168.61.25	192.168.60.1	DNS	92	Standard query 0x0ch7 A _en-undater.brave.com

Explore and understand color-coded packet markings.



Name	Filter
☒ Bad TCP	tcp.analysis.flags && !tcp.analysis.window_update && !tcp.analysis.keep_alive && !tcp.analysis.keep_alive_ack
☒ HSRP State Change	hsrp.state != 8 && hsrp.state != 16
☒ Spanning Tree Topology Change	stp.type == 0x80
☒ OSPF State Change	ospf.msg != 1
☒ ICMP errors	icmp.type in { 3, 5, 11 } icmpv6.type in { 1, 4 }
☒ ARP	arp
☒ ICMP	icmp icmpv6
☒ TCP RST	tcp.flags.reset eq 1
☒ SCTP ABORT	sctp.chunk_type eq ABORT
☒ IPv4 TTL low or unexpected	(ip.dst != 224.0.0.0/4 && ip.ttl < 5 && !(pim ospf eigrp bgp tcp.port == 179)) (ip.dst == 224.0.0.0/24 && ip.dst != 224.0.0.251 && ip.ttl != 1 &&
☒ IPv6 hop limit low or unexpected	(ipv6.dst != ff00::/8 && ipv6.hlim < 5 && !(ospf bgp tcp.port == 179)) (ipv6.dst == ff00::/8 && ipv6.hlim not in { 1, 64, 255 })
☒ Checksum Errors	eth.fcs.status == "Bad" ip.checksum.status == "Bad" tcp.checksum.status == "Bad" udp.checksum.status == "Bad" sctp.checksum.status == "Bad" ms
☒ SMB	smb nbss nbns netbios
☒ HTTP	http tcp.port == 80 http2
☒ DCERPC	dcerpc
☒ Routing	hsrp eigrp ospf bgp cdp vrrp carp gvrp igmp ismp
☒ TCP SYN/FIN	tcp.flags & 0x02 tcp.flags.fin == 1
☒ TCP	tcp
☒ UDP	udp
☒ Broadcast	eth[0] & 1
☒ System Event	systemd_journal sysdig

Analyze the handshake process in Wireshark

The TCP three-way handshake is the process used by the Transmission Control Protocol (TCP) to establish a reliable connection between a client and a server before data transfer begins.

Filtering Only Handshake Packets

SYN (Synchronize)

SYN-ACK (Synchronize + Acknowledge)

ACK (Acknowledge)

Only SYN Packets

tcp.flags.syn == 1 && tcp.flags.ack == 0						
No.	Time	Source	Destination	Protocol	Length	Info
56	1.961933	192.168.61.25	192.168.60.1	TCP	66	63706 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
57	1.962099	192.168.61.25	192.168.60.1	TCP	66	61101 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
75	1.971731	192.168.61.25	172.67.68.93	TCP	66	61165 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
248	6.565497	192.168.61.25	192.168.60.1	TCP	66	62392 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
249	6.565609	192.168.61.25	192.168.60.1	TCP	66	62482 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
269	6.612797	192.168.61.25	192.178.211.84	TCP	66	57375 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
428	9.818763	192.168.60.229	192.168.61.145	TCP	66	51776 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
437	9.980990	192.168.61.25	23.193.165.65	TCP	66	57376 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
496	10.828559	192.168.60.229	192.168.61.145	TCP	66	[TCP Retransmission] 51776 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
547	12.844079	192.168.60.229	192.168.61.145	TCP	66	[TCP Retransmission] 51776 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
662	16.859414	192.168.60.229	192.168.61.145	TCP	66	[TCP Retransmission] 51776 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
949	29.191594	192.168.61.40	192.168.61.145	TCP	66	59268 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
957	30.205025	192.168.61.40	192.168.61.145	TCP	66	[TCP Retransmission] 59268 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
994	32.210354	192.168.61.40	192.168.61.145	TCP	66	[TCP Retransmission] 59268 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1152	36.224772	192.168.61.40	192.168.61.145	TCP	66	[TCP Retransmission] 59268 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1160	36.348430	192.168.61.25	192.168.60.1	TCP	66	56078 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1161	36.348524	192.168.61.25	192.168.60.1	TCP	66	60614 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1177	36.350701	192.168.61.25	142.250.70.106	TCP	66	62176 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1288	39.957203	192.168.61.25	8.8.8.8	TCP	66	63012 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1296	39.980083	192.168.61.25	8.8.8.8	TCP	66	60391 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1304	40.071285	192.168.61.25	23.193.165.43	TCP	66	60392 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1515	49.490977	192.168.61.25	192.168.60.1	TCP	66	53988 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1516	49.491328	192.168.61.25	192.168.60.1	TCP	66	58327 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1534	49.503785	192.168.61.25	192.168.60.1	TCP	66	50668 → 8090 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1897	60.011982	192.168.61.25	57.144.43.32	TCP	66	59602 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

SYN-ACK Packets

tcp.flags.syn == 1 && tcp.flags.ack == 1						
No.	Time	Source	Destination	Protocol	Length	Info
58	1.962823	192.168.60.1	192.168.61.25	TCP	66	53 → 63706 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
59	1.962823	192.168.60.1	192.168.61.25	TCP	66	53 → 61101 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1	2.001711	172.67.68.93	192.168.61.25	TCP	66	443 → 61165 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8192
250	6.566401	192.168.60.1	192.168.61.25	TCP	66	53 → 62482 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
251	6.566401	192.168.60.1	192.168.61.25	TCP	66	53 → 62392 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
272	6.629371	192.178.211.84	192.168.61.25	TCP	66	443 → 57375 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
438	10.011355	23.193.165.65	192.168.61.25	TCP	66	80 → 57376 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1162	36.349105	192.168.60.1	192.168.61.25	TCP	66	53 → 56078 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1163	36.349105	192.168.60.1	192.168.61.25	TCP	66	53 → 60614 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1183	36.366824	142.250.70.106	192.168.61.25	TCP	66	443 → 62176 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1289	39.987106	8.8.8.8	192.168.61.25	TCP	66	443 → 63012 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1297	40.019175	8.8.8.8	192.168.61.25	TCP	66	443 → 60391 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1305	40.087204	23.193.165.43	192.168.61.25	TCP	66	80 → 60392 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1517	49.491946	192.168.60.1	192.168.61.25	TCP	66	53 → 53988 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1518	49.491946	192.168.60.1	192.168.61.25	TCP	66	53 → 58327 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1535	49.504703	192.168.60.1	192.168.61.25	TCP	66	8090 → 50668 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
1899	60.041152	57.144.43.32	192.168.61.25	TCP	66	443 → 59602 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM WS=256
1900	60.041152	57.144.43.32	192.168.61.25	TCP	66	5222 → 60269 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM WS=256
1988	61.124622	57.144.43.32	192.168.61.25	TCP	66	443 → 56957 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM WS=256
2063	62.417000	103.182.221.100	192.168.61.25	TCP	66	443 → 57090 [SYN, ACK] Seq=0 Ack=1 Win=15312 Len=0 MSS=1392 SACK_PERM WS=256
2568	69.589166	192.168.60.1	192.168.61.25	TCP	66	53 → 56703 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
2569	69.589166	192.168.60.1	192.168.61.25	TCP	66	53 → 56239 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
2597	69.898529	44.254.57.201	192.168.61.25	TCP	66	443 → 52487 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460 SACK_PERM WS=256
2612	70.169708	44.254.57.201	192.168.61.25	TCP	66	443 → 57606 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460 SACK_PERM WS=256
2616	70.175878	23.193.165.43	192.168.61.25	TCP	66	80 → 57607 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128

ACK Packets

tcp.flags.ack == 1 && tcp.len == 0						
No.	Time	Source	Destination	Protocol	Length	Info
316	6.944780	192.178.211.84	192.168.61.25	TCP	60	443 → 57375 [ACK] Seq=5995 Ack=11104 Win=259328 Len=0
332	8.079488	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=9936 Ack=106 Win=228 Len=0
337	8.704918	192.168.61.25	20.42.73.26	TCP	54	65127 → 443 [FIN, ACK] Seq=1 Ack=1 Win=1019 Len=0
347	8.860549	192.168.61.25	104.18.32.47	TCP	54	52460 → 443 [ACK] Seq=106 Ack=10114 Win=253 Len=0
368	8.902975	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10114 Ack=1506 Win=228 Len=0
369	8.902975	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10114 Ack=4306 Win=229 Len=0
370	8.902975	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10114 Ack=7690 Win=230 Len=0
371	8.902975	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10114 Ack=8195 Win=230 Len=0
391	8.954211	20.42.73.26	192.168.61.25	TCP	60	443 → 65127 [FIN, ACK] Seq=1 Ack=2 Win=16385 Len=0
392	8.954285	192.168.61.25	20.42.73.26	TCP	54	65127 → 443 [ACK] Seq=2 Ack=2 Win=1019 Len=0
396	9.056029	192.168.61.25	172.64.148.235	TCP	54	61930 → 443 [ACK] Seq=1 Ack=155 Win=254 Len=0
400	9.074564	192.168.61.25	104.18.32.47	TCP	54	52460 → 443 [ACK] Seq=8195 Ack=10533 Win=251 Len=0
438	10.011355	23.193.165.65	192.168.61.25	TCP	66	80 → 57376 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
439	10.011493	192.168.61.25	23.193.165.65	TCP	54	57376 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
441	10.013296	23.193.165.65	192.168.61.25	TCP	60	80 → 57376 [ACK] Seq=1 Ack=112 Win=64128 Len=0
444	10.044141	23.193.165.65	192.168.61.25	TCP	60	80 → 57376 [FIN, ACK] Seq=188 Ack=112 Win=64256 Len=0
445	10.044244	192.168.61.25	23.193.165.65	TCP	54	57376 → 80 [ACK] Seq=112 Ack=189 Win=65280 Len=0
446	10.044420	192.168.61.25	23.193.165.65	TCP	54	57376 → 80 [FIN, ACK] Seq=112 Ack=189 Win=65280 Len=0
447	10.073728	23.193.165.65	192.168.61.25	TCP	60	80 → 57376 [ACK] Seq=189 Ack=113 Win=64256 Len=0
461	10.728245	172.64.148.235	192.168.61.25	TCP	60	443 → 61930 [ACK] Seq=155 Ack=92 Win=19 Len=0
482	10.773646	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10533 Ack=10995 Win=231 Len=0
483	10.773646	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10533 Ack=15799 Win=232 Len=0
484	10.773646	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10533 Ack=15934 Win=232 Len=0
485	10.775844	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10533 Ack=17334 Win=233 Len=0
486	10.775844	104.18.32.47	192.168.61.25	TCP	60	443 → 52460 [ACK] Seq=10533 Ack=23567 Win=234 Len=0